

30.7.26

C02-AT3-PEER REVIEW

1. How to add a New node at the beginning

Logic:

1. Allocate memory for the node.
2. Store the data.
3. Make the next pointer NULL.
4. Attach the node to the list.

Code:

void at the begin()

{

struct node * temp;

temp = (struct node *) malloc (sizeof (struct node));

printf ("Enter node data");

scanf ("%d", & temp -> data);

temp -> left = NULL;

temp -> right = NULL;

if (root == NULL)

{

root = temp;

}

else

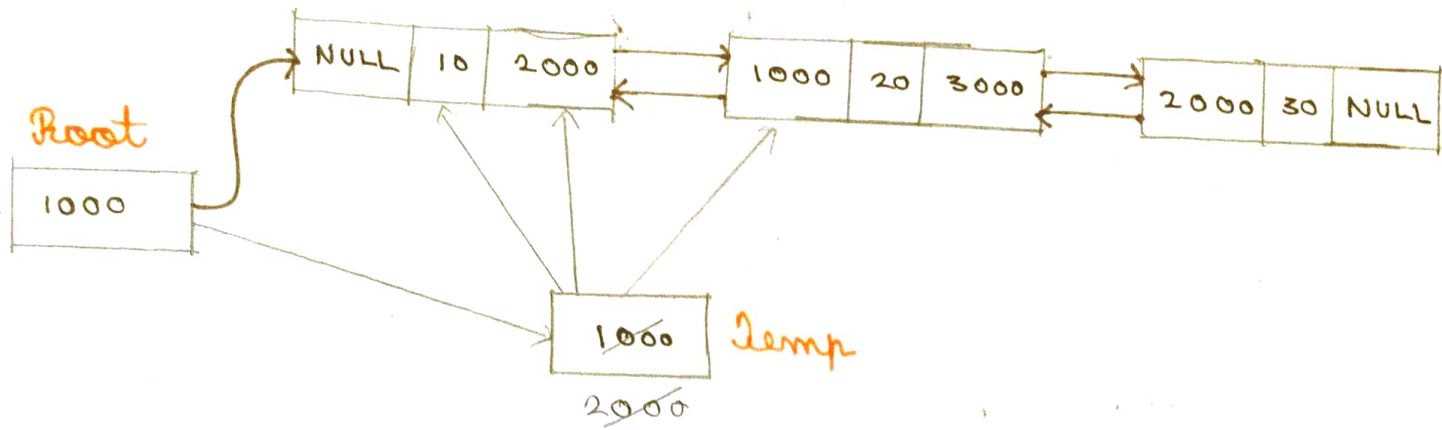
temp -> right = root;

root -> left = temp;

root = temp;

}

Eg:



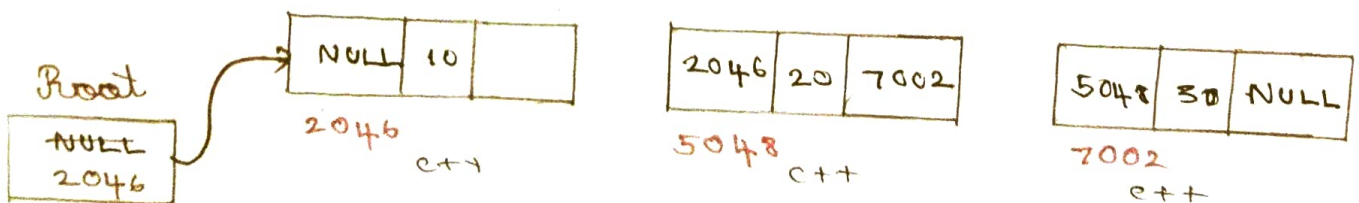
Find the length:

Algorithm:

1. Start from the head node.
2. Initialize count = 0.
3. Traverse the list until NULL.
4. Increment count for every node.
5. Print count.

Code:

```
int length()
{
    struct node * temp = root;
    int count = 0;
    while (temp != NULL)
    {
        count++;
        temp = temp -> right;
    }
    return count;
}
```



Display / Print the Data:

```
void display()
```

```
{
```

```
    struct node * temp = root;
```

```
    while (temp != NULL)
```

```
    {
```

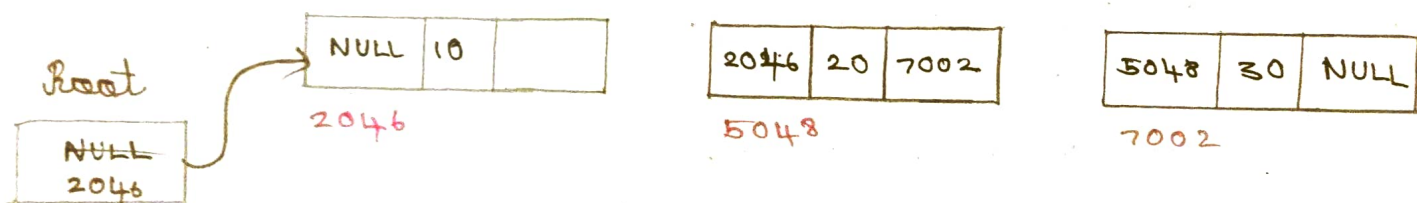
```
        printf("%d →", temp->data);
```

```
        temp = temp->right;
```

```
    }
```

Algorithm:

1. Start from the head node.
2. If head == NULL, print "List is Empty".
3. Otherwise, move through each node.
4. Print the data of each node.
5. Stop when temp == NULL.



Traversal:

temp → 10 (Print 10)

temp → 20 (Print 20)

temp → 30 (Print 30)

temp → NULL (stop).

2. Tower of Hanoi:

Algorithm:

1. If there is only 1 disk, move it from Source to destination.
2. Move $n-1$ disks from Source to Auxiliary.
3. Move the largest disk from Source to Destination.
4. Move $n-1$ disks from Auxiliary to Destination.

Code:

```
void TOH (int n, char S, char A, char D)
{
    if (n == 1)
    {
        printf ("Move disk 1 from %c to %c\n", S, D);
        return;
    }
    TOH (n-1, S, D, A);
    printf ("Move disk %d from %c to %c\n", n, S, D);
    TOH (n-1, A, S, D);
}
```

Eg:

Initial State:

Source (A)	Aux (B)	Dest (C)
1		
2		
3		

Step 1: Move disk 2 from A $\rightarrow$ B

Source (A)	Aux (B)	Dest (C)
2		1
3		

Step 2: Move Disk 2 from A $\rightarrow$ B

Source (A)	Disk (B)	Dest (C)
3	2	1

Step 3: Move Disk 1 from C $\rightarrow$ B

Source (A)	Disk (B)	Dest (C)
3	1 2	

Step 4: Move Disk 3 from A $\rightarrow$ C

Source (A)	Disk (B)	Dest (C)
	1 2	3

Step 5: Move Disk 1 from B $\rightarrow$ A

Source (A)	Disk (B)	Dest (C)
1	2	3

Step 6: Move Disk 2 from B $\rightarrow$ C

Source (A)	Disk (B)	Dest (C)
1		2 3

Step 7: Move Disk 1 from A $\rightarrow$ C

Source (A)	Disk (B)	Dest (C)
		1 2 3